MONASH University

FIT5128– MASTER'S THESIS

FlowLaCAM*: Pre-emptive Congestion-Aware Neural Guidance for LaCAM*

Topic: Learning-Enhanced Strategies in Multi-Agent Path Finding

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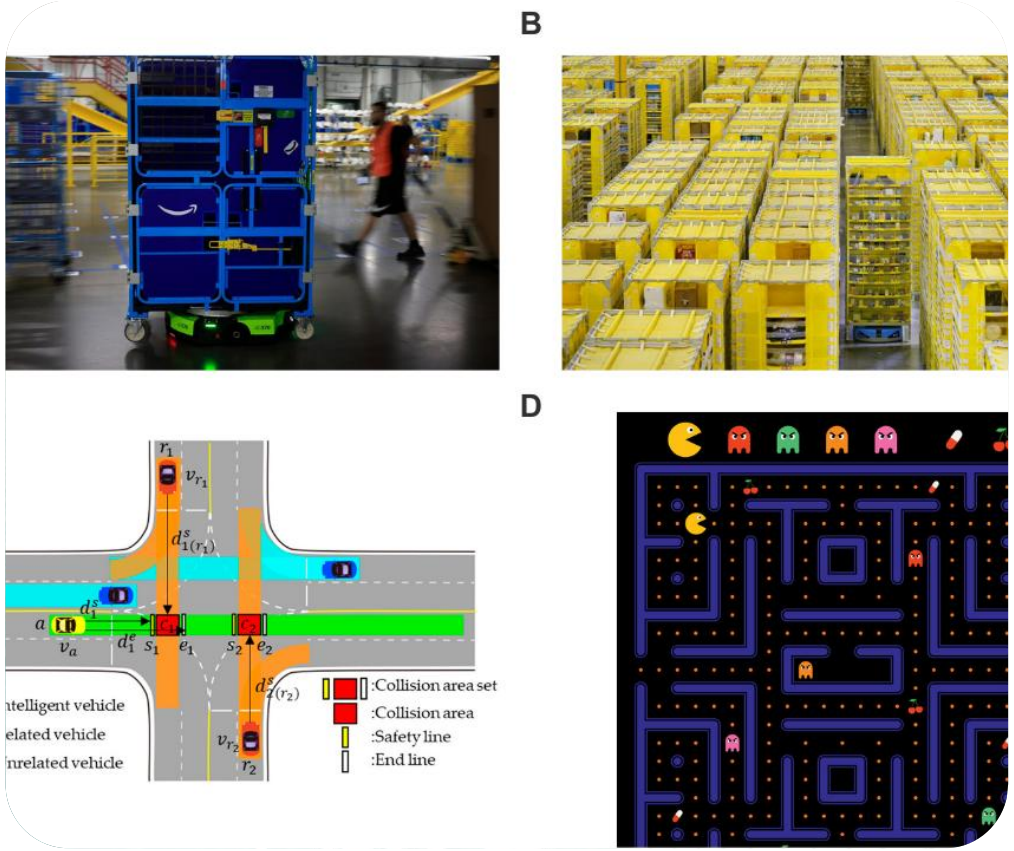
OVERVIEW OF MULTI-AGENT PATH FINDING (MAPF) AND ITS APPLICATIONS



Sourced from 1, 2. Figure shows the real-world application of MAPF in Amazon Fulfillment centers, traffic signals, and games like PacMan

What is MAPF?

Multi-Agent Path Finding (MAPF) is the coordination of multiple autonomous agents, ensuring each reaches its individual goal in a shared environment without collisions or deadlocks.

Why is MAPF Important?

Enables large-scale automation in:

- Warehouse robotics (e.g., moving shelves and goods)
- Autonomous vehicle fleets and traffic management
- Gaming & crowd simulation (coordinated movement of many characters)
- Critical for safety, efficiency, and system throughput in modern intelligent environments

Evaluation Metrics:

In MAPF, we evaluate solution quality using three metrics — **Sum of Costs**, which measures the total path length of all agents; **Makespan**, the time until the last agent reaches its goal; and **Sum of Loss**, which quantifies deviation from the optimal baseline.

MAPF SOLUTION PARADIGMS

Also called Expert Algorithms

CLASSICAL MAPF SOLVERS

Heuristic Search based: Finds paths using heuristic estimates to guide agents towards goals efficiently. [A* (Hart et al., 1968), ODrM* (Walker et al., 2018), and LaCAM* (Okumura et al., 2023)]

Conflict-based: Resolves agent conflicts hierarchically, ensuring collision-free joint solutions. [CBS (Sharon et al., 2015), ECBS (Barer et al., 2014)]

Rule-based: Coordinates agents with priority and backtracking rules for decentralized planning. [PIBT (Okumura et al., 2022)]

Neighbourhood Search-based: Improves paths by locally re-optimizing agent groups across neighbourhoods. [MAPF-LNS (Li et al., 2021)]

ML-based

LEARNING-BASED SOLVERS

Imitation Learning based: Trains agents to mimic expert planners or demonstrations for scalable path finding. [PRIMAL (Sartoretti et al., 2019), SCRIMP (Li et al., 2023)]

Reinforcement Learning based: Uses trial and error with rewards to teach agents efficient coordination and collision avoidance. [PRIMAL2(Sartoretti et al., 2021)]

Graph Neural Network based: Learns agent policies using neural networks that capture inter-agent connectivity and map structure. [GNN-MAPF(Li et al., 2020), MAGAT (Wang et al., 2020)]

Expert + Learning based

HYBRID SOLVERS

Integrate learning modules with classical MAPF solvers to leverage adaptability and structural guarantees for scalable, high-quality planning.

[Neural LNS (Ma et al., 2021), MAPF-ML-LNS (Li et al., 2022), FOLLOWER (Tolmidis & Tzafestas, 2023), SSIL(Sartoretti et al., 2020), Online Guidance Graph Optimization (Zang et al., 2025) , FlowLaCAM*]

Thesis focus

CORE PROBLEM/GAP IN MAPF

No MAPF solution is simultaneously scalable, fast, and high-quality!

Hybrid solvers represent the most promising direction to achieve it

Scalable: Solvers must efficiently manage many agents and large, complex environments.

CBS, ECBS struggle to scale due to exponential growth in conflict resolution or coordination overhead

Fast: Algorithms must provide quick solutions for real-time control or efficiently improve plans as time allows.

Anytime methods like PIBT and LaCAM* can produce rapid initial solutions but may degrade in crowded scenarios; pure learning models (e.g., PRIMAL) can return fast but less reliable paths.

High Quality Solution: Must deliver high-quality, coordinated joint plans

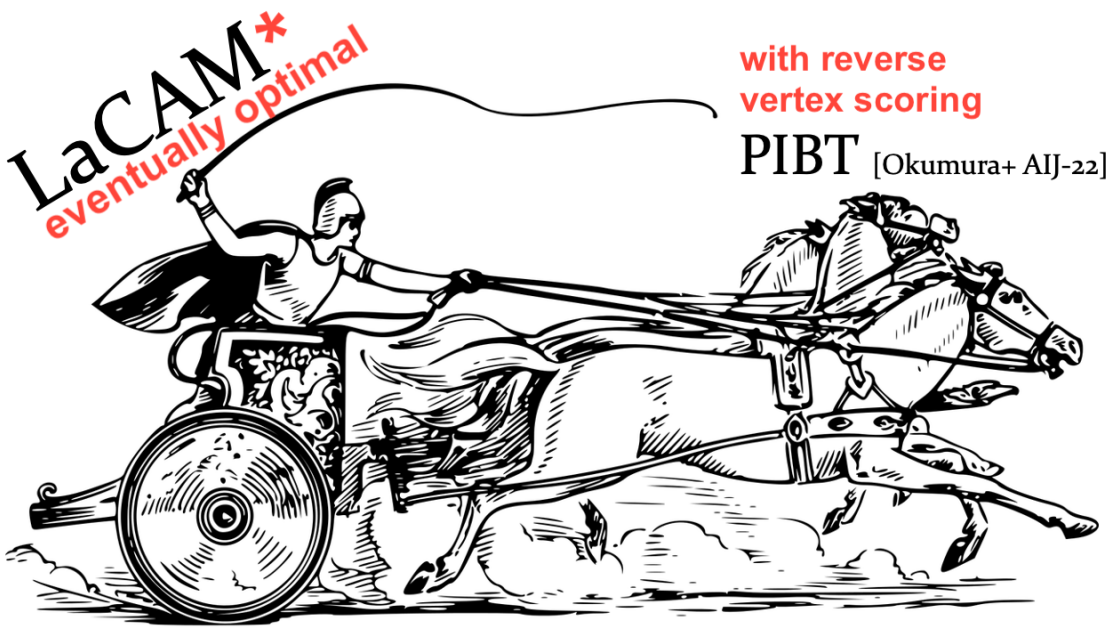
Expert algorithms like CBS, LNS2 trade speed for higher solution quality.

RELATED WORK – LaCAM* AND PIBT

LaCAM* AND PIBT

LaCAM* *(Okumura, 2024)*

- **Anytime two-level framework** combining joint-state search and local conflict resolution.
- Uses **lazy reverse BFS** from the goal to expand and refine joint plans.
- Resolves collisions via **PIBT's priority exchange**, ensuring **completeness** and **incremental optimality**.
- Employs **re-parenting** to dynamically improve solution quality and convergence speed.



Sourced from link. The author explains LaCAM* and PIBT's relationship through this graphic.

PIBT (Okumura et al., 2022)

- Rule-based framework using **dynamic priority exchange** to resolve conflicts.
- Agents move **sequentially by priority**, with lower-priority agents replanning when conflicts occur.
- Ensures **deadlock-free** and **fair** coordination through local decision-making.
- Highly **efficient** but lacks **global coordination**, leading to deadlocks and suboptimal paths.

Limitation

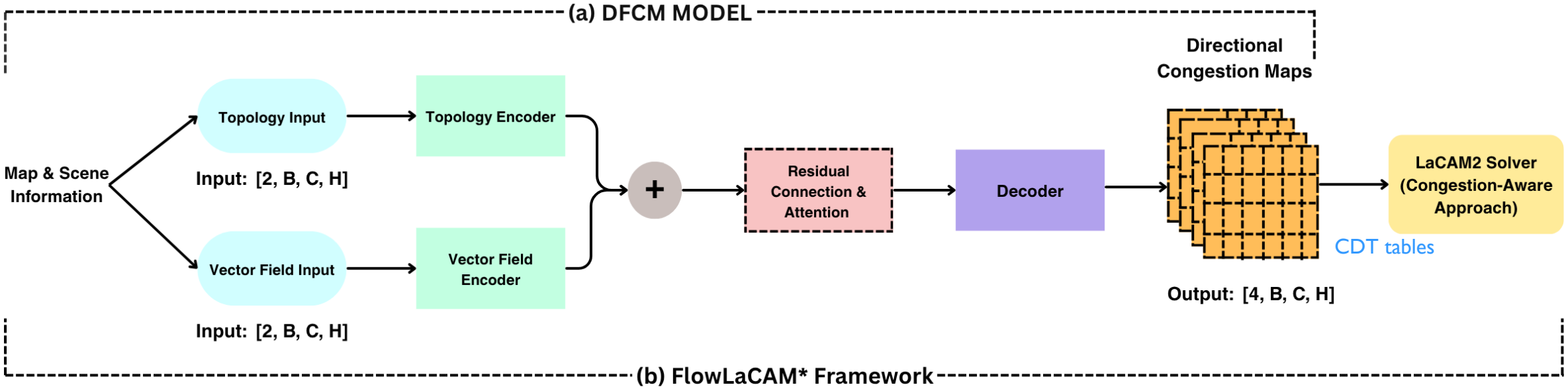
As noted by *Okumura (2024), "the initial solution quality is far from optimal, and its convergence speed to the optimum is slow,"* highlighting the framework's **inefficiency in early-stage planning**.

- Heuristic *bias toward individual shortest paths*, ignoring congestion and coordination.
- Causes *bottleneck overcommitment and slow convergence*.

PROPOSED SOLUTION – FLOWLaCAM*

FLOWLaCAM*

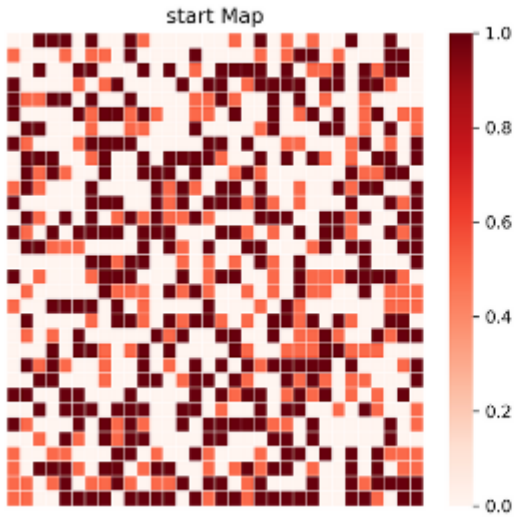
- Core idea: Anticipates congestion before search begins, much like how navigation systems use live traffic data to avoid bottlenecks.
- Integrates learned congestion priors from the Directional Flow Congestion Model (DFCM) into LaCAM*.
- DFCM outputs Directional Congestion Matrices (DCMs) that quantify traversal costs for each movement direction.
- The DCMs are then converted into a Congestion Distance Table (CDT), bridging the neural model and search process.
- The CDT acts as a persistent congestion-aware cost map, guiding agent coordination throughout the anytime search.

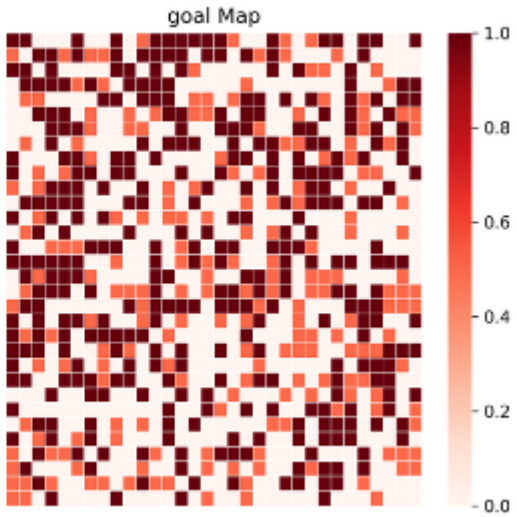


FlowLaCAM* framework architecture

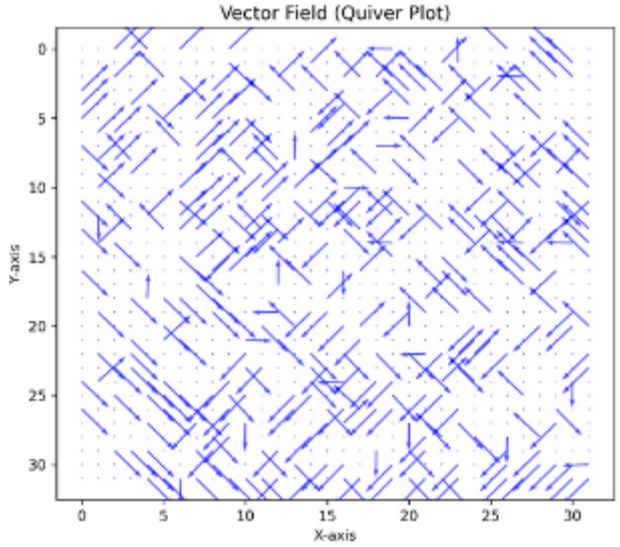
FLOWLaCAM*- Inputs and Outputs

Topology Inputs – Start and Goal Maps

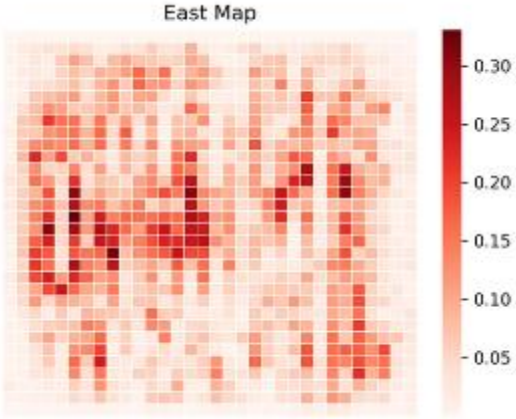


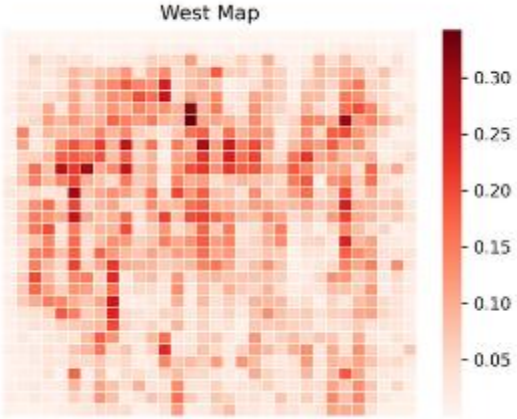


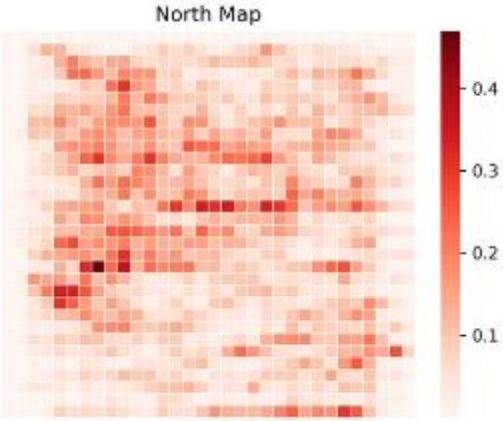
Vector Field Inputs

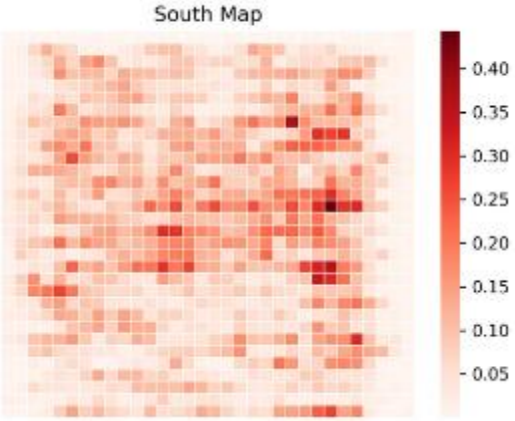


Pre-emptive Directional Congestion Maps

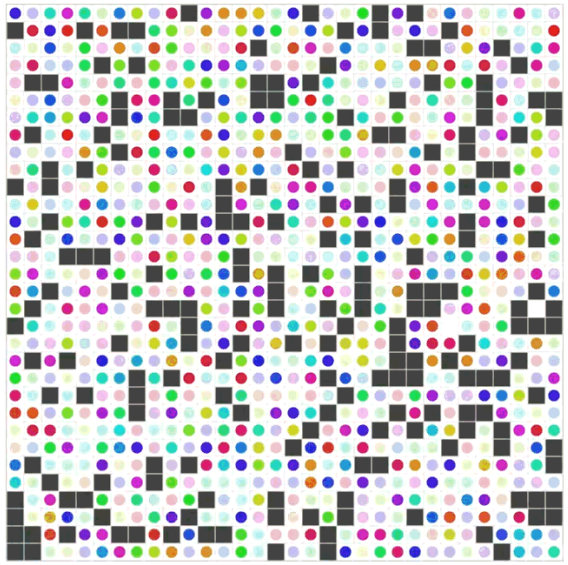


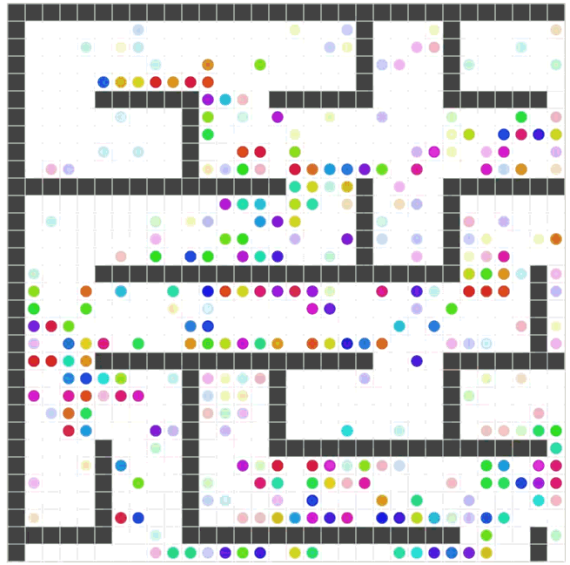


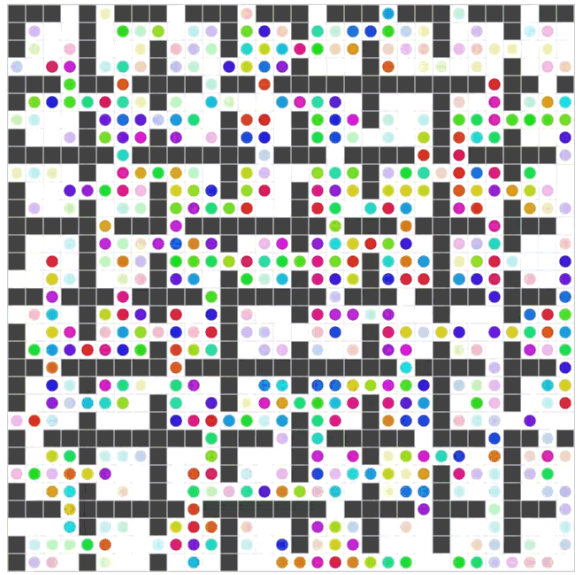




RESULTS







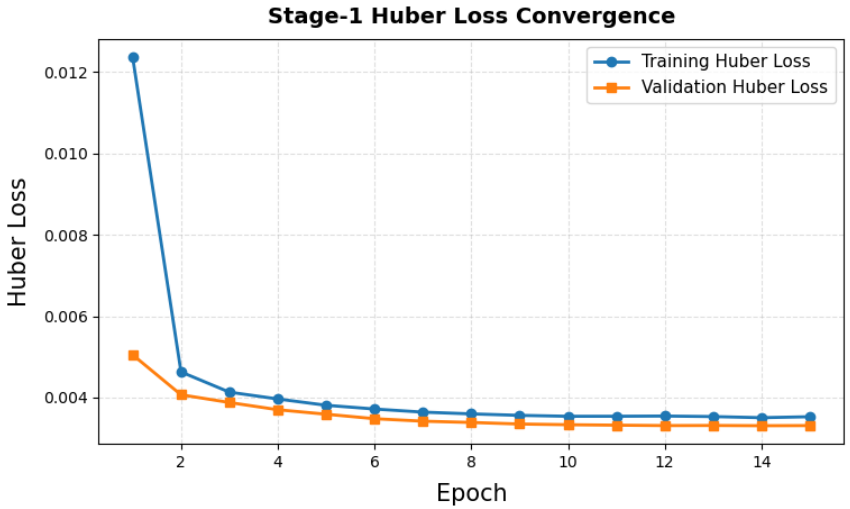
TRAINING RESULTS

Training Procedure: FlowLaCAM* training was done in two stages (1&2)

Stage 1: Learn **directional congestion values** from expert trajectories using the **DFCM** (Direction Flow Congestion Model).

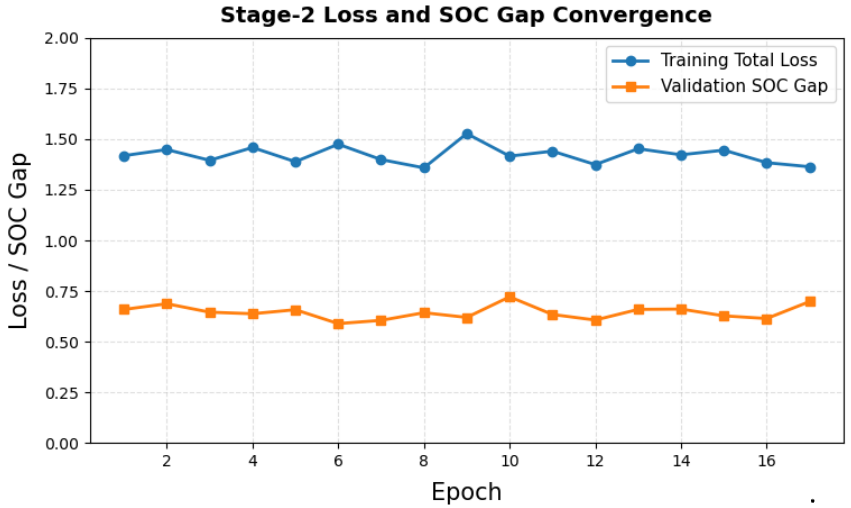
Stage 2: Fine-tune FlowLaCAM* to optimise the **Sum of Costs (SOC)**, a key MAPF evaluation metric.

DFCM Training Results



Smooth convergence of **Huber Loss**, confirming stable learning of congestion priors.

FlowLaCAM* Training Results



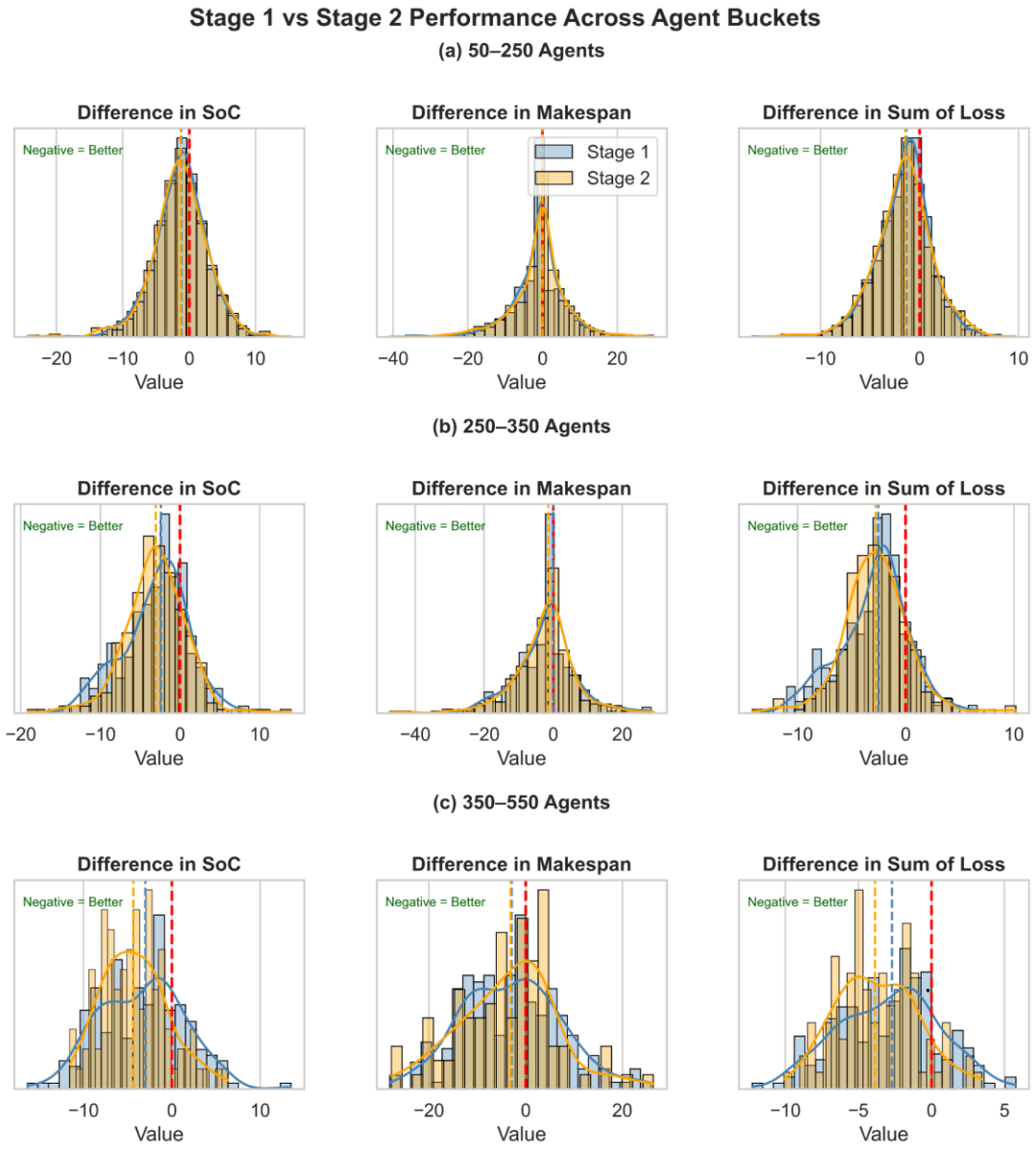
Consistent **Loss–SOC gap** trend, indicating generalisation and improved planning efficiency.

COMPARATIVE ANALYSIS: STAGE-1 AND STAGE-2 PERFORMANCE

- **Stage 2 consistently outperforms Stage 1** across all agent densities, showing lower Sum of Costs and cumulative loss with reduced variance — confirming improved efficiency and stability.

- **Performance gains strengthen beyond 250 agents**, highlighting better coordination under congestion and stronger generalisation of learned flow patterns.

- Makespan improvements, though subtler, indicate smoother temporal synchronisation and **fewer prolonged completion times — signaling enhanced cooperative timing**.



COMPARATIVE WITH BASELINE - LACAM*

- **Consistent Gains Across Environments:** FlowLaCAM* achieved 1–5% lower Sum of Costs, 1–2% lower Makespan, and 2–4% fewer Wait Steps, with ~75% of runs outperforming LaCAM*.

- **Stronger Under Congestion:** The largest gains (up to 5–8%) appeared in random, high-density maps, showing better congestion handling and cooperative flow.

- **High Success Rate and Stability:** FlowLaCAM* maintained 100% success rate in five out of six environments, and only minor drops in obstacle-free maps (92% and 69%) — confirming robust convergence and planner reliability.

Table 1: Map-wise performance summary of FlowLaCAM* model across agent count buckets. For each metric, the table reports mean Δ, standard deviation (Std), and percentage of instances improved (%Imp). Negative Δ indicates improvement over the baseline (LaCAM*). N denotes sample size.

Map	Agents	Δ Sum of Costs			Δ Makespan			Δ Total Wait Steps			N
		Mean	Std	%Imp	Mean	Std	%Imp	Mean	Std	%Imp	
maze-32-32-4	50–149	−1.80	5.27	70.7	−0.35	9.83	46.6	−2.48	3.23	82.0	887
	150–299	−1.94	9.67	67.1	0.23	16.12	50.5	−3.29	5.66	76.8	495
maze-32-32-2	50–149	−0.45	2.84	57.6	−0.50	4.38	48.8	−0.74	2.24	63.8	735
	150–299	−2.15	3.92	72.2	−1.89	5.97	60.3	−2.00	2.97	77.1	345
room-32-32-4	50–149	−0.78	4.13	57.3	−1.64	7.48	52.9	−0.35	3.09	54.0	770
	150–299	−1.34	4.73	63.0	−0.67	6.67	52.1	−1.04	3.59	62.8	651
	300–499	−1.95	4.74	64.8	−2.11	7.90	63.0	−1.79	3.49	69.1	165
empty-32-32	50–149	−0.72	3.20	59.6	0.29	1.66	7.8	−0.95	1.56	71.0	900
	150–299	−1.37	2.26	73.6	0.20	2.41	20.0	−1.83	1.66	87.0	1358
	300–499	−2.75	2.50	87.0	0.12	3.47	32.9	−2.89	1.97	**93.2**	529
random-32-32-20	50–149	−1.17	3.70	62.7	0.17	4.85	31.8	−1.26	2.55	68.5	705
	150–299	−3.95	4.06	82.9	−4.65	8.13	65.4	−3.30	3.09	85.5	674
	300–499	**-5.10**	4.76	84.4	**-5.28**	8.71	71.9	−4.12	3.64	86.6	494
random-32-32-10	50–149	−0.49	3.01	55.3	−0.03	2.31	19.6	−0.96	1.96	68.0	685
	150–299	−2.86	2.78	85.5	−1.95	4.93	51.1	−2.76	2.31	89.0	675
	300–499	−4.21	3.34	89.0	**-5.09**	7.54	71.4	−3.70	2.68	**91.8**	535

Table 2: Map-wise success rate of FlowLaCAM* summary. N denotes the total number of evaluated instances per map. In all scenarios, the baseline LaCAM* had a 100% success rate.

Map	Number of Instances (N)	Success Rate (%)
empty-32-32	5014	69.21
maze-32-32-2	1308	100.00
maze-32-32-4	1772	100.00
random-32-32-10	2715	92.01
random-32-32-20	2485	100.00
room-32-32-4	2165	100.00

FUTURE WORK & CONCLUSION

- FlowLaCAM* introduced a hybrid learning–search approach, integrating the Directional Flow Congestion Model (DFCM) into LaCAM*, allowing learned congestion priors to influence heuristic reasoning directly.
- The framework achieved consistent improvements in Sum of Costs, Makespan, and Waiting Steps, especially under dense and stochastic conditions, while maintaining solver completeness and stability.
- These results highlight that embedding learned congestion awareness can enhance coordination quality without sacrificing efficiency or scalability.
- However, current congestion priors are static, limiting adaptability in dynamic settings and low-structure maps.
- Future research should focus on adaptive congestion re-estimation, topology-aware scaling, and cross-solver integration to extend DFCM's generalisability and robustness.
- Ultimately, this work marks a step toward scalable, interpretable, and deployable multi-agent coordination guided by learning-informed heuristics.

THANK YOU!

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